

Signal Processing for Beyond-5G and 6G Communications

Waveforms ■ ultra-massive MIMO ■ RIS ■ ISAC ■ AI-native PHY

Dr. Markkandan S

Associate Professor

Invited talk ■ 90 minutes ■ 2026

School of Electronics Engineering

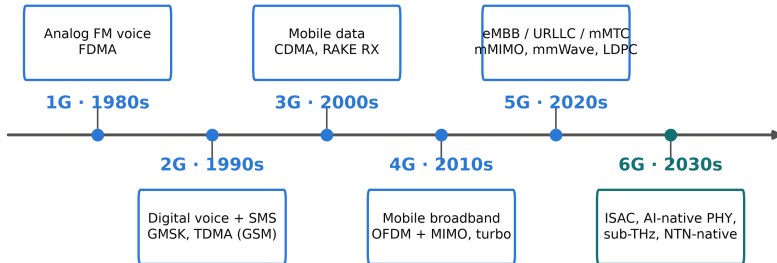
Vellore Institute of Technology, Chennai

`markkandan.s@vit.ac.in`

Opening & Motivation

Six generations in forty years

Every generation is defined by its physical-layer signal processing



- Each decade: a new generation, a 10–100× peak-rate jump
- The differentiator each time was the **physical-layer signal processing choice**:
FDMA → TDMA → CDMA → OFDM/MIMO → mMIMO

Extended from 5G:

- Immersive communication
- Hyper-reliable & low-latency
- Massive communication

Genuinely new for 6G:

- Integrated sensing & communication
- AI and communication
- Ubiquitous connectivity

Design values

Sustainability ▪ security & resilience ▪ connecting the unconnected ▪ interoperability

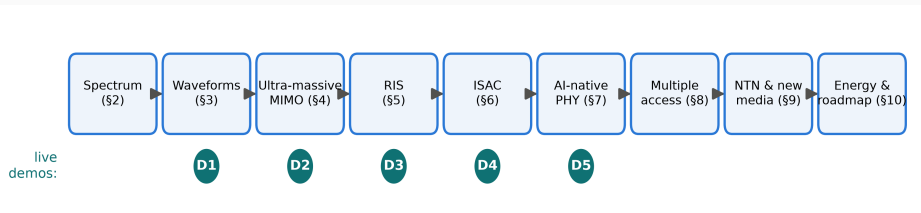
KPI targets: IMT-2020 vs IMT-2030

KPI	5G (IMT-2020)	6G (IMT-2030, research)
Peak data rate	20 Gb/s	200 Gb/s – 1 Tb/s
User-plane latency	1 ms	0.1 – 1 ms
Connection density	$10^6/\text{km}^2$	$10^6 - 10^8/\text{km}^2$
Positioning	m-level	cm-level (+ velocity)
Sensing	—	native KPI class
Energy	qualitative	first-class design value

- IMT-2030 deliberately gives *ranges*: targets are scenario-dependent, not one number for everything

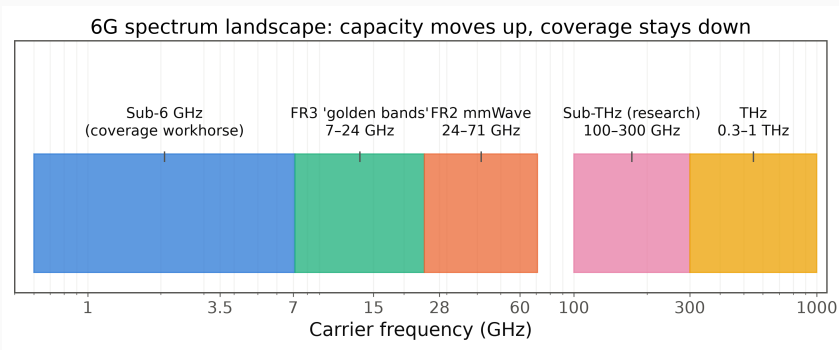
Why signal processing is the bottleneck *and* the enabler

- At sub-6 GHz we operate close to Shannon: coding, MIMO, scheduling are mature
⇒ gains must come from **new dimensions**
- New dimensions on offer: bandwidth (THz), aperture (ultra-massive arrays), *environment* (RIS), *function* (sensing), *semantics* (AI)
- Hardware gets *worse* as we go up in frequency — DSP must compensate (impairment-aware design)



New Spectrum Frontiers

The 6G spectrum landscape

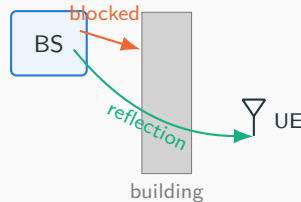


- Sub-6: coverage workhorse, congested
- FR2 mmWave: deployed, niche
- **FR3 (7–24 GHz)**: the pragmatic 6G capacity band; WRC-23 identified new IMT bands (e.g. around 4.4–4.8, 7.1–8.4, 14.8–15.35 GHz, region-dependent)
- Sub-THz 100–300 GHz: the research frontier

- 7–15 GHz: $2\text{--}4\times$ more bandwidth than C-band with *manageable* path loss — the coverage–capacity compromise
- Same physical aperture $\Rightarrow 4\text{--}9\times$ more antenna elements than at 3.5 GHz: massive MIMO scales *naturally*
- Coexistence with incumbents (satellite, radar) $\Rightarrow$ sharing and sensing become part of the PHY job
- DSP price: wider bandwidth $\times$ bigger arrays = beam squint, calibration, channel-estimation burden

mmWave: what a decade taught us

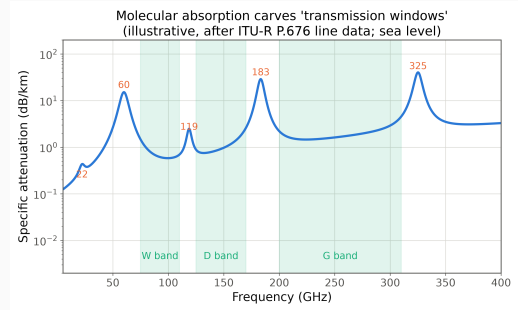
- FR2 works — but blockage (hand/body 10–25 dB), rain of small cells, beam management overhead limited rollout
- Killer app became fixed wireless access, not mobility
- **Lesson 1:** link-budget math wins arguments
- **Lesson 2:** beam training overhead is a *rate* tax



Sub-THz channels: 100–300 GHz

$$PL(f, d) = \underbrace{\left(\frac{4\pi fd}{c}\right)^2}_{\text{spreading}} \cdot \underbrace{e^{\kappa_{\text{abs}}(f)d}}_{\text{molecular abs.}}$$

- O₂/H₂O resonances at 60, 119, 183, 325 GHz carve **transmission windows**
- Channel: sparse, specular, LoS-dominated
- Consequences: short links, pencil beams mandatory



Hardware impairments dominate at high bands

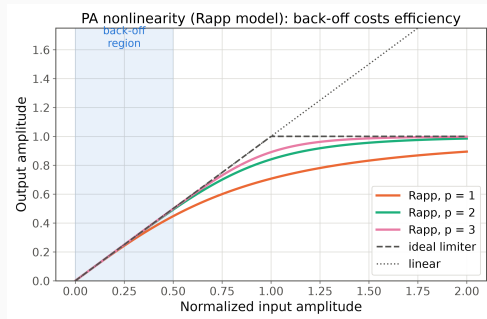
Rapp PA model (AM/AM):

$$g(A) = \frac{GA}{\left[1 + \left(\frac{GA}{A_{\text{sat}}}\right)^{2p}\right]^{1/2p}}$$

Wiener phase noise:

$$\varphi[n+1] = \varphi[n] + w[n], \quad w \sim \mathcal{N}(0, 4\pi^2 f_c^2 c T_s)$$

- PA efficiency falls with f ; back-off $\Rightarrow$ PAPR is a first-class metric
- Phase noise power $\propto f_c^2$; low-resolution ADCs to save power



Implications for waveform design

Requirement	sub-6	FR3	mmWave	sub-THz
Low PAPR	nice	nice	want	need
Phase-noise robustness	—	—	want	need
Doppler robustness	HST	HST	want	want
Spectral confinement	want	want	nice	nice
MIMO compatibility	need	need	need	want

- No single waveform wins every column $\Rightarrow$ expect **scenario-dependent waveform choices** in 6G
- CP-OFDM's weak spots are exactly the columns that harden at high bands

Waveforms Beyond OFDM

CP-OFDM recap: why it won 4G and 5G

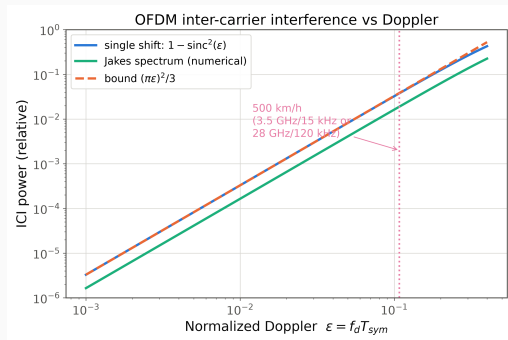
$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}, \quad \text{CP: linear conv.} \rightarrow \text{circular conv.} \rightarrow \text{1-tap FDE}$$

- One FFT turns a frequency-selective channel into N flat channels
- MIMO-friendly (per-subcarrier processing), flexible numerology, gigantic ecosystem
- Hidden assumption: **channel constant over one symbol** (quasi-static)
- That assumption is precisely what high mobility and high carriers break

Where OFDM breaks: high Doppler, high bands

$$P_{\text{ICI}} \lesssim \frac{(\pi f_d T_s)^2}{3}$$

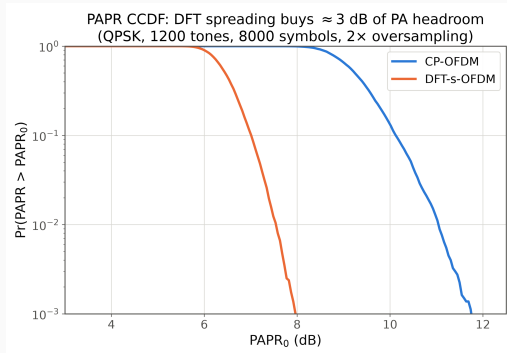
- Doppler spread destroys orthogonality $\Rightarrow$ ICI
- 500 km/h at 28 GHz: $f_d \approx 13$ kHz ($\epsilon \approx 0.11$ even at 120 kHz spacing)
- PAPR ~ 10 – 12 dB $\Rightarrow$ PA back-off — worst fit for THz PAs
- Phase noise adds CPE + ICI on top



DFT-s-OFDM and the single-carrier comeback

$$\mathbf{X} = F_M \mathbf{s} \quad (\text{DFT spread})$$

- Quasi single-carrier envelope: ≈ 3 dB PAPR advantage
- Already the 5G uplink workhorse — proven at scale
- Sub-THz proposals: SC-FDE blocks; equalizer complexity moves to RX
- Leading candidate for 6G high bands in early 3GPP discussion



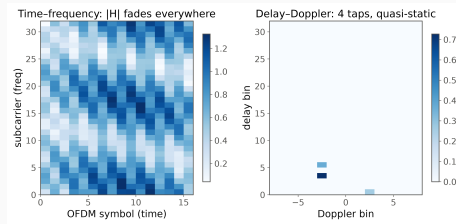
Filtered multicarrier: lessons of the 5G bake-off

- Candidates circa 2016: FBMC, UFMC, GFDM, f-OFDM — better spectral confinement, relaxed synchronicity
- Why they lost: MIMO integration pain (FBMC/OQAM), complexity, marginal gains once windowing was allowed
- What survived: spectral shaping as a *transmitter implementation choice* below the standard
- **Meta-lesson:** ecosystem inertia is a design constraint — a radical waveform needs **radical** gains, not 0.5 dB

OTFS I: thinking in delay–Doppler

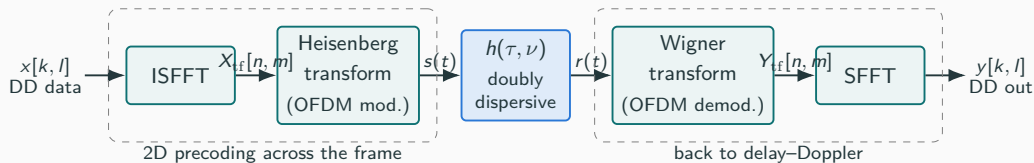
$$r(t) = \iint h(\tau, \nu) s(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu$$

- In delay–Doppler (DD) the channel is **sparse and quasi-static**: a few (τ_i, ν_i) taps = physical reflectors with range & velocity
- Each DD symbol rides the *entire* TF channel $\Rightarrow$ full-diversity potential



OTFS II: transceiver structure

$$X_{\text{tf}}[n, m] = \frac{1}{\sqrt{NM}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] e^{j2\pi\left(\frac{nk}{N} - \frac{ml}{M}\right)} \quad (\text{ISFFT})$$



- OTFS = 2D pre-/post-processing wrapped around an OFDM modem $\Rightarrow$ attractive migration path

OTFS III: input–output relation and detection

$$y[k, l] \approx \sum_i h_i x[(k - k_i)_N, (l - l_i)_M] e^{j2\pi\phi_i} + w[k, l]$$

- A 2D twisted circular convolution: each tap shifts data in delay and Doppler and adds a phase
- Sparsity $\Rightarrow$ message-passing / low-complexity MMSE detectors
- Costs: frame-level latency, detection complexity, fractional-Doppler leakage
- Status check (honest): rich literature, strong results — *not yet* in 3GPP normative work

AFDM: the chirp-based challenger

$$\phi_m[n] = \frac{1}{\sqrt{N}} e^{j2\pi\left(c_1 n^2 + \frac{mn}{N} + c_2 m^2\right)} \quad (\text{discrete affine Fourier basis})$$

- Multiplex on **chirps**; tune c_1 to the expected Doppler span $\Rightarrow$ all paths separate in the DAFT domain
- Full diversity in doubly dispersive channels with *1D, single-symbol* processing (vs OTFS frame-level)
- FFT-based implementation; rising fast in the literature since 2023
- Open: MIMO integration, pilot design, robustness studies at scale

Waveform scorecard

Waveform scorecard (5 = best; author's assessment)

CP-OFDM	1	2	5	5	5
DFT-s-OFDM	4	2	4	4	5
SC-FDE	5	3	3	3	3
OTFS	2	5	2	2	1
AFDM	3	5	3	2	1
	Low PAPR	Doppler robustness	Low RX complexity	MIMO maturity	Standards readiness

► **Live demo 1 (3.5 min) runs here**

OFDM vs OTFS at 500 km/h — watch the BER floor appear on screen, live.

matlab/demo1_ofdm_vs_otfs

Ultra-Massive MIMO & Beamforming

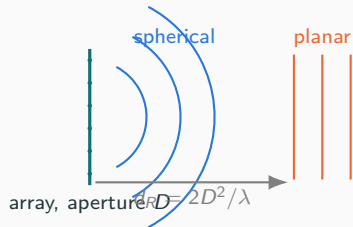
From massive to ultra-massive

- 5G today: 64 TRX at 3.5 GHz $\rightarrow$ 6G: 512–4096 elements at FR3 / mmWave / sub-THz
- Same rooftop aperture at higher $f \Rightarrow$ more elements, narrower beams, more EIRP
- New physics enters: near-field propagation, spatial non-stationarity across large arrays
- The bill: RF chains, ADCs, calibration, fronthaul — DSP must economize

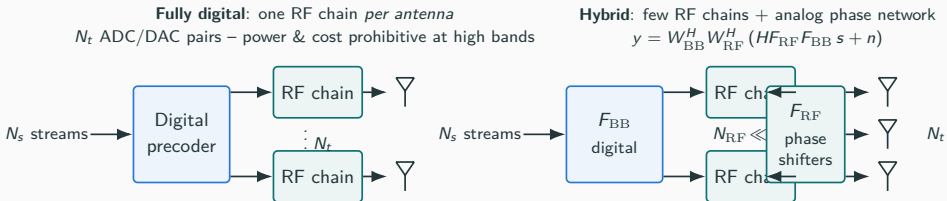
Near-field vs far-field

$$d_R = \frac{2D^2}{\lambda} \quad \begin{cases} d > d_R : & \text{planar, } \mathbf{a}(\theta) \\ d < d_R : & \text{spherical, } \mathbf{a}(\theta, r) \end{cases}$$

- $D = 0.5$ m at 30 GHz: $d_R = 50$ m — users are *inside* the near field
- Near field $\Rightarrow$ beam focusing (range-selective!), not just steering
- Wavefront curvature carries range information — gift for localization



Hybrid analog–digital architectures

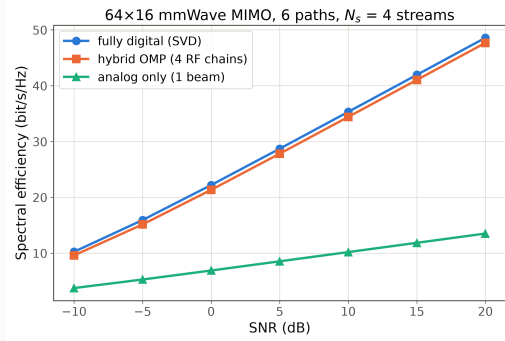


- F_{RF} : phase-only, $|[F_{\text{RF}}]_{ij}| = 1/\sqrt{N_t}$; wideband arrays add **true-time-delay** units against beam squint

Designing the hybrid precoder: sparsity to the rescue

$$\min_{F_{\text{RF}}, F_{\text{BB}}} \|F_{\text{opt}} - F_{\text{RF}} F_{\text{BB}}\|_F \quad \text{s.t.} \quad F_{\text{RF}} \in \{\text{array response vectors}\}$$

- mmWave channels are sparse in angle
⇒ precoding $\approx$ sparse reconstruction
⇒ OMP (El Ayach et al. 2014)
- $N_{\text{RF}} \approx N_s$ chains already $\approx$ near-digital performance

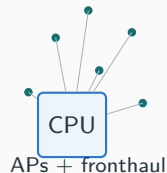
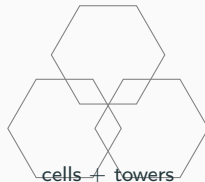


Beam management: finding the needle

- 5G NR baseline: SSB sweep $\rightarrow$ CSI-RS refinement $\rightarrow$ tracking and failure recovery
- Overhead grows with codebook size $\Rightarrow$ hierarchical search, ML-assisted prediction for 6G
- Sub-THz pencil beams: mobility and micro-blockage stress the loop hard
- Research directions: side-information beams (position, vision, radar), near-field codebooks (angle $\times$ range)

Cell-free massive MIMO

- Many distributed low-cost APs, coherent joint processing, no cell boundaries
- Uniform SNR replaces cell-edge misery; macro-diversity beats blockage
- The catches: fronthaul, synchronization, scalable processing (dynamic cooperation clusters)
- 6G angle: natural platform for **distributed sensing** and RIS integration



Channel estimation by compressed sensing

$$\mathbf{y} = \Phi \mathbf{h} + \mathbf{n}, \quad \|\mathbf{h}\|_0 = L \ll \dim(\mathbf{h}), \quad H \approx A_R \text{diag}(\mathbf{g}) A_T^H$$

- Angular-domain sparsity (L paths) $\Rightarrow$ recovery from $\mathcal{O}(L \log N)$ pilots via OMP / AMP
- Pilot savings are what make 1024-element arrays *trainable at all*
- Caveats: grid mismatch (off-grid/atomic-norm methods), near-field dictionaries grow with $\text{angle} \times \text{range}$
- Same math returns in Section 8 for activity detection — learn once, bill twice

$$\text{SINR}_j \xrightarrow{M \rightarrow \infty} \frac{\beta_{jj}^2}{\sum_{l \neq j} \beta_{jl}^2} \quad (\text{same-pilot cells } l)$$

- Finite coherence block $\Rightarrow$ pilot reuse $\Rightarrow$ correlated estimates $\Rightarrow$ coherent interference that does **not** average out with more antennas
- Modern view (Björnson et al.): with multi-cell MMSE processing and spatial correlation, capacity grows without bound — contamination is a *processing artifact*, not a law of nature
- Practical mitigations: smart pilot assignment, correlation-aware estimation, cell-free operation

Array frontier: open challenges

- Wideband beam **squint** at 10%+ fractional bandwidth: TTD-augmented hybrids
- Near-field CSI: dimensionality, codebooks, tracking
- Spatial non-stationarity: different subarrays see different channels
- The honest metric: **energy per beamformed bit**, not array gain

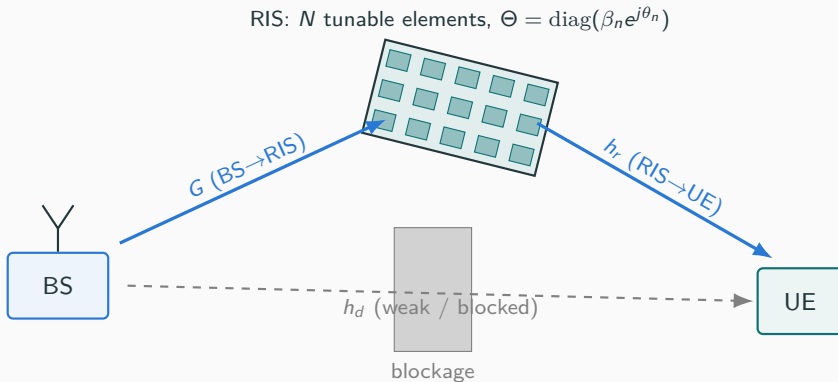
► **Live demo 2 (3.5 min) runs here**

64-element array, live: steer the beam with a slider; digital vs OMP-hybrid spectral efficiency.

`matlab/demo2_hybrid_beamforming`

Reconfigurable Intelligent Surfaces

RIS: programming the channel itself



$$y = (\mathbf{h}_r^H \Theta \mathbf{G} + h_d^H) x + n \quad \text{path loss} \propto (d_1 d_2)^{-\alpha} \text{ (dyadic)}$$

The cascaded channel model

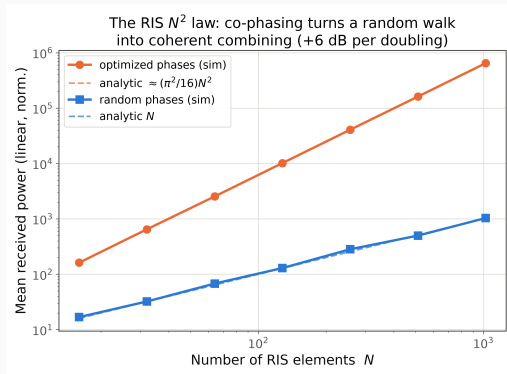
$$y = \left(\mathbf{h}_r^H \Theta \mathbf{G} + h_d^H \right) x + n, \quad \Theta = \text{diag}(\beta_1 e^{j\theta_1}, \dots, \beta_N e^{j\theta_N})$$

- Product path loss $\propto (d_1 d_2)^{-\alpha}$: the **fundamental tax of passivity** — both hops paid in full
- Estimation burden: cascaded coefficients scale as $N \times M$; structure (sparsity, common subspaces) to the rescue
- Large N at short range $\Rightarrow$ the RIS itself operates in the near field

Phase optimization and the N^2 law

$$\theta_n^* = \arg(h_d) - \arg([\mathbf{h}_r]_n^* [\mathbf{G}]_n) \Rightarrow P_{\text{rx}} \propto \left(\sum_n |[\mathbf{h}_r]_n| |[\mathbf{G}]_n| \right)^2 \sim N^2$$

- Co-phasing: amplitude $\propto N \Rightarrow$ power $\propto N^2$ (+6 dB per doubling)
- Random phases: 2D random walk $\Rightarrow$ power $\propto N$
- Multi-user / MIMO: non-convex $\Rightarrow$ SDR, manifold, alternating optimization



$$y = \left(\mathbf{h}_r^H \Theta \mathbf{G} \right) x + \underbrace{\mathbf{h}_r^H \Theta \mathbf{n}_{\text{RIS}}}_{\text{amplified noise}} + n$$

- Passive gain often eaten by dyadic loss $\Rightarrow$ **active RIS**: reflection with amplification — and an amplified-noise tax
- Variants: STAR-RIS (transmit *and* reflect), beyond-diagonal RIS, holographic surfaces
- The uncomfortable benchmark: does an active RIS beat a plain relay / small cell at the same cost and power? Must be argued per scenario

Deployment: where RIS actually helps

- Sweet spots: mmWave/sub-THz **blockage bypass**, indoor dead zones, localized capacity boosts
- Placement rule of thumb: near TX or near RX — mid-path maximizes the dyadic loss penalty
- Overheads that gate practicality: channel estimation and the control link, both scaling with N
- Field trials (metro, industrial) encouraging; standardization pre-normative (ETSI ISG RIS; 3GPP interest)

- CSI acquisition with no RF chains on the panel
- Wideband behavior: one phase per element, many subcarriers
- EM-consistent models: mutual coupling, non-ideal reflection
- RIS-aided sensing and localization: surfaces as programmable landmarks

► **Live demo 3 (2.5 min) runs here**

The N^2 law, live: SNR gain from 16 to 1024 elements, random vs optimized phases.

matlab/demo3_ris_gain

Integrated Sensing & Communication

- One spectrum, one infrastructure, **two functions**: communication + radar-style sensing
- Drivers: spectrum scarcity; a dense cellular grid is a ready-made sensor network; XR/V2X/drones need perception
- IMT-2030 lists ISAC as a *defining* scenario — not an add-on
- Economic pitch: sensing “for free” on waveforms you already transmit

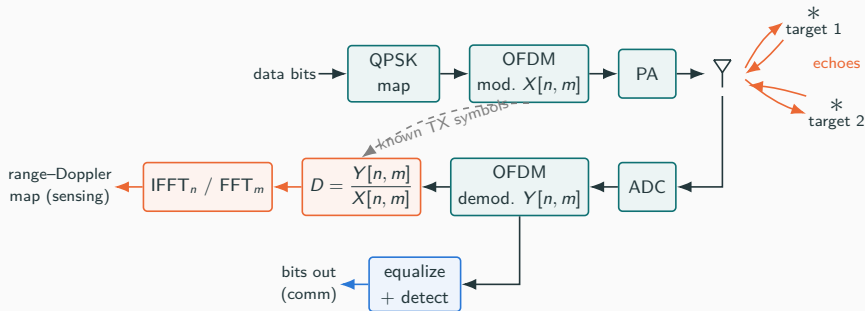
Radar refresher in one slide

$$\tau = \frac{2R}{c}, \quad f_D = \frac{2vf_c}{c}, \quad \Delta R = \frac{c}{2B}, \quad \Delta v = \frac{c}{2f_c T_{\text{obs}}}$$

- Bandwidth buys range resolution; *coherent observation time* buys velocity resolution
- Matched filtering + coherent integration = the SNR workhorses; ambiguity function encodes the trade-offs
- All classic DSP — 6G reuses it wholesale, on comm hardware

OFDM as a radar waveform

$$D[n, m] = \frac{Y[n, m]}{X[n, m]} \xrightarrow[\text{FFT over } m]{\text{IFFT over } n} \text{range-Doppler map}$$



Joint waveform design

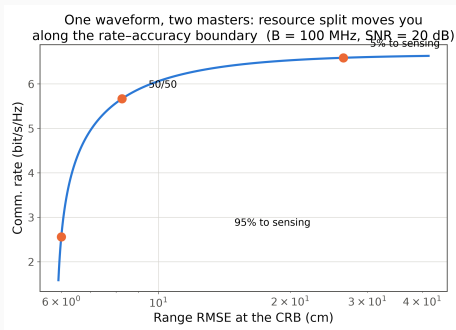
- Design axes: power/subcarrier allocation, dedicated sensing symbols vs full data reuse, beampattern shaping
- Inherent friction: comm wants **randomness** (capacity), radar wants **determinism** (clean ambiguity function)
- Recent theory: the “random-data radar” penalty is quantifiable — and often acceptably small
- Sensing-assisted communication closes the loop: radar tracks feed beam prediction

Fundamental limits: CRB vs rate

$$\text{var}(\hat{\tau}) \geq \frac{1}{8\pi^2 \beta_{\text{rms}}^2 \text{SNR} \cdot N_{\text{obs}}}$$

(β_{rms} : RMS bandwidth)

- Delay CRB: RMS bandwidth is king — edge subcarriers are gold for ranging
- Rate-CRB region: the capacity-distortion picture of ISAC
- Waterfilling for rate $\neq$ waterfilling for sensing $\Rightarrow$ principled compromises



Monostatic, bistatic, and network sensing

- **Monostatic:** full waveform knowledge; but TX–RX **self-interference** is brutal (full duplex or quasi-CW operation)
- **Bistatic** (BS→UE, BS→BS): no self-interference; synchronization and clock offsets bite instead
- **Network/multistatic:** fusion across cells — geometry diversity, coverage; cellular timing infrastructure helps



- Rel-19: dedicated [channel-model study for ISAC](#) (targets with RCS enter the stochastic model); SA1 use cases feed RAN
- Rel-20 (6G study): sensing among the candidate native features; ETSI ISG ISAC in parallel
- Honest status: pre-normative — first commercial “sensing” will likely be positioning-class network features, then micro-Doppler applications

► [Live demo 4 \(3 min\) runs here](#)

Two moving targets on a range–Doppler map — computed from a frame that is simultaneously carrying QPSK data.

`matlab/demo4_isac_range_doppler`

AI-Native Physical Layer

What “AI-native” actually means

- 5G: AI bolted on (SON, scheduling). 6G ambition: AI as a **design principle** inside the air interface
- **Model deficit**: no faithful math model (hardware nonlinearities, mmWave clutter) $\Rightarrow$ learn the model
- **Algorithm deficit**: model known, optimal algorithm intractable $\Rightarrow$ learn the algorithm
- Integration spectrum: replace a block $\rightarrow$ co-design blocks $\rightarrow$ end-to-end learned PHY

- Treat the pilot-grid estimate as a **noisy image** $\Rightarrow$ CNN denoising / super-resolution across time–frequency
- Learned estimators approach MMSE *without* knowing channel statistics — they absorb them from data
- Gains largest at low SNR and under model mismatch
- Costs: training data, generalization across cells/hardware, μ s-latency inference budgets

Autoencoder end-to-end PHY

- O'Shea & Hoydis 2017: TX-channel-RX as one autoencoder; learn constellation and demapper **jointly**
- Learned constellations rediscover — and under impairments beat — hand-designed ones
- Needs a differentiable channel $\Rightarrow$ surrogate models or RL for the real world



- Replace equalization + demapping with one trained network over the received grid (channel estimate optional!)
- Demonstrated at 3GPP-compatible scale (multi-user MIMO neural receivers, Sionna-class toolchains)
- Robustness from training over channel *distributions*; complexity tamed by pruning/quantization
- Open operational question: retraining cadence vs channel drift

Model-based deep learning: deep unfolding

$$\text{ISTA: } \mathbf{x}^{(k+1)} = \eta_{\lambda} \left(\mathbf{x}^{(k)} + A^H (\mathbf{y} - A \mathbf{x}^{(k)}) \right) \Rightarrow \text{LISTA: } \mathbf{x}^{(k+1)} = \eta_{\lambda_k} \left(W_1^{(k)} \mathbf{y} + W_2^{(k)} \mathbf{x}^{(k)} \right)$$

- Unroll K iterations of a classical algorithm into a K -layer network; **learn the knobs** (steps, thresholds, matrices)
- Order-of-magnitude fewer iterations at equal accuracy; retains interpretability and per-layer guarantees flavor
- PHY wins: unfolded detection, beamforming, sparse channel estimation

- Shift the target from **bit fidelity** to **task fidelity**: transmit what the receiver needs to act, not every bit
- Neural joint source–channel coding: graceful degradation, no cliff effect — compelling image/speech demos
- Open theory: a working semantic information measure; where the separation theorem stands; standardizable interfaces
- Fair positioning: promising for constrained links (XR, satellite IoT); not a 6G day-one feature

The hard problems: data, generalization, trust

- **Data:** channel measurements are vendor-siloed; sim-to-real transfer is the quiet blocker
- **Generalization:** across cells, hardware, mobility; distribution-shift monitoring in a live network
- **Budget:** μ s-scale inference, joules per inference, PHY real-time constraints
- **Trust:** testability, certification, explainability — telecom-grade requirements
- Community opening: standardized channel datasets & benchmarks would be a genuine service to the field

- Rel-18 study → Rel-19 normative first steps; three use cases: **CSI feedback compression, beam prediction, positioning**
- Two-sided models (UE encoder / network decoder): the interoperability puzzle of the decade
- 6G study (Rel-20): AI-native candidates; the real spec work is *life-cycle management* — monitoring, fallback, retraining

► **Live demo 5 (2.5 min) runs here**

LS vs MMSE vs a small neural network for OFDM channel estimation — NMSE curves, pre-trained net, runs in seconds.

`matlab/demo5_dl_channel_estimation`

Multiple Access & Spectrum Sharing

Beyond orthogonal access

- OMA (orthogonal slices) is capacity-optimal only in corner cases; 6G stress: massive connectivity + wildly heterogeneous QoS
- Landscape: power-domain NOMA, code-domain (SCMA), RSMA, grant-free access
- 5G reality check: NOMA studied (MUST), *not* widely deployed — SIC complexity vs modest gains
- Reframe: multiple access = an interference-management philosophy: avoid, cancel, or partially decode

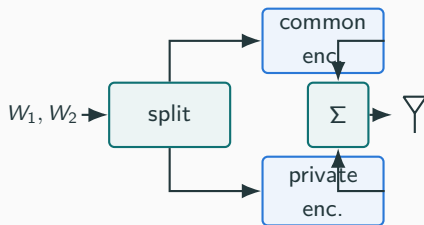
$$R_{\text{far}} = \log_2 \left(1 + \frac{P_2 |h_f|^2}{P_1 |h_f|^2 + \sigma^2} \right), \quad R_{\text{near}} = \log_2 \left(1 + \frac{P_1 |h_n|^2}{\sigma^2} \right)$$

- Superpose users in power; near user decodes far user's signal, **cancels**, then decodes its own
- Gains hinge on channel disparity ($|h_n| \gg |h_f|$)
- Taxes: SIC error propagation, CSI sensitivity, scheduling complexity — degrades ungracefully with mobility

Rate-splitting multiple access (RSMA)

$$R_k = C_k + R_{p,k}, \quad \sum_k C_k \leq \min_k \log_2(1 + \text{SINR}_{c,k})$$

- Split each message: **common part** (decoded by all) + **private part** (only own)
- One SIC step: decode common, subtract, decode private
- Spans SDMA $\leftrightarrow$ NOMA $\leftrightarrow$ multicast as special cases of one framework



Grant-free and unsourced random access

- mMTC regime: short packets, sporadic activity, signaling handshakes cost more than the payload
- Activity detection = **compressed sensing** again: few active users out of many $\Rightarrow$ sparse recovery / covariance methods
- Unsourced RA: all users share one codebook; decoder returns an unordered message list — clean info-theoretic framing
- Same sparsity toolbox as Section 4 — third billing for the same math

Why RSMA keeps gaining traction for 6G

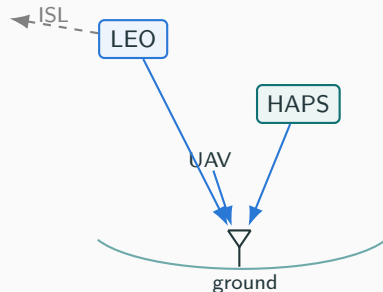
- One framework unifying the access zoo — fewer special cases to standardize
- Graceful degradation under **imperfect CSIT** (the regime real networks live in)
- Documented synergies with multi-antenna ISAC and RIS beamforming
- Entry fee: encoder/scheduler complexity, common-rate control signaling

“Split, superpose — and let SIC do bounded work.”

Coverage from the Sky & New Media

Non-terrestrial networks: the 3D architecture

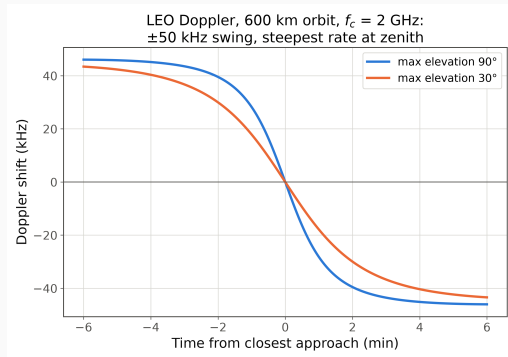
- Layers: LEO mega-constellations (300–1200 km), HAPS (~20 km), UAV relays
- 6G goal: one 3GPP-native fabric — common waveform family, seamless sky↔ground mobility
- Direct-to-device NTN is already commercially real; scaling it is the 6G task



The LEO signal processing problem: Doppler

$$f_D(t) = \frac{f_c}{c} v_{\text{rel}}(t), \quad v_{\text{sat}} \approx 7.6 \text{ km/s} \Rightarrow \pm 25 \text{ ppm } (\pm 50 \text{ kHz at 2 GHz})$$

- Doppler *rate* up to $\sim \text{kHz/s}$ near zenith; RTT 2–25 ms breaks HARQ timing
- The gift: **ephemeris is known** $\Rightarrow$ pre-compensate, track only the residual
- DD-domain waveforms (Section 3) are natural fits here



NTN in 3GPP: status in one minute

- Rel-17: NR-NTN baseline (transparent payloads, timing/Doppler handling)
- Rel-18/19: coverage & mobility enhancements, IoT-NTN, regenerative payloads on the way
- 6G: NTN native from the start — inter-satellite routing, store-and-forward IoT, unified mobility

- IM/DD constraint: real, non-negative signals $\Rightarrow$ DCO-/ACO-OFDM adaptations of familiar math
- LiFi: dense indoor capacity, security by physics (light stays in the room); FSO: fiber-class backhaul without the trench — incl. ground-satellite
- Challenges: alignment, ambient light, the uplink asymmetry
- Honest niche: industrial floors, cabins, RF-denied environments

Quantum-secured links (brief)

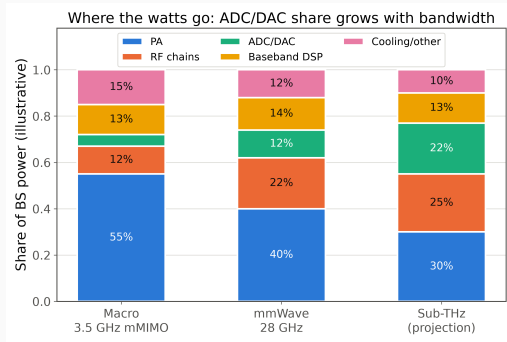
- QKD: key distribution whose security rests on physics; fiber and satellite demos exist
- Realism: point-to-point, modest key rates $\Rightarrow$ practical path is **hybrid** QKD + post-quantum cryptography
- For 6G: primarily a security-architecture topic, not a PHY-DSP one — scoped to one slide on purpose

Energy, Standardization & Roadmap

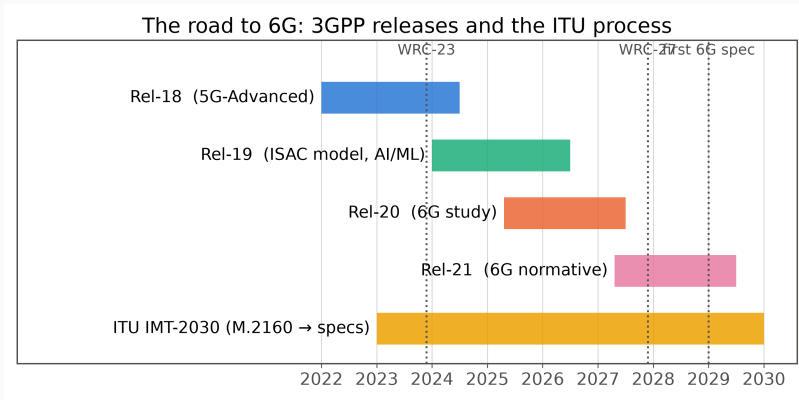
Energy-efficient PHY design

$$\eta_{EE} = \frac{R}{P_{\text{tot}}} \text{ [bit/J]}, \quad P_{\text{ADC}} \propto 2^b f_s$$

- Network energy: OPEX + sustainability KPI (an IMT-2030 design value)
- Levers: PA operating point (waveform!), low-resolution ADCs, sleep-aware signal design, lean always-on signals
- Design shift: bit/J next to bit/s/Hz



The 3GPP road to 6G



- Rel-19 carries the studies that matter here (ISAC model, AI/ML); Rel-20 opens the 6G study; Rel-21 goes normative $\Rightarrow$ first spec $\approx$ 2028–29

- M.2160 framework (2023) → requirements & evaluation methodology (2024–26)
→ proposal submissions → IMT-2030 specifications $\approx$ 2029–30
- Same machinery that turned IMT-2020 into “5G” — history rhymes
- Feeding programs: Hexa-X-II (EU), Next G Alliance (US), IMT-2030 Promotion Group (CN), Bharat 6G (IN)

A research roadmap for this room

Start now (paper-sized):

- DD-domain receivers & AFDM studies
- Near-field CSI & codebooks
- ISAC waveform trade-offs (CRB-rate)
- Deep-unfolded PHY algorithms

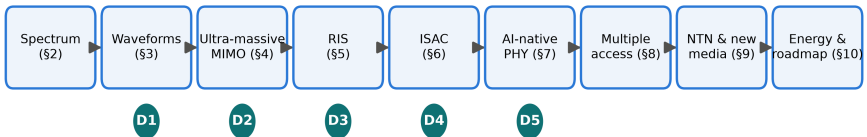
Build toward (project-sized):

- Cell-free + RIS testbeds & field data
- AI life-cycle management for PHY
- Sub-THz prototyping & measurements
- Public channel datasets & benchmarks

Choose problems where **DSP insight beats brute force** — that is our comparative advantage.

The 6G signal-processing map

live
demos:



- Recurring themes: sparsity ▪ domain transforms ▪ hardware-aware design ▪ model + data hybrids

Takeaways & Q&A

Five demos, five lessons

1. **D1** OFDM vs OTFS: at high Doppler, **change the domain**, not the SNR
2. **D2** Hybrid beamforming: in sparse channels, **4 RF chains $\approx$ 64**
3. **D3** RIS: co-phasing turns a random walk into **N^2 combining**
4. **D4** ISAC: a **data-carrying frame** can image two moving targets
5. **D5** Learned estimation: **$\approx$ MMSE without knowing the statistics**

Pre-computed versions: appendix A1–A5 ■ code shipped with the slides

Key takeaways

- 6G = new spectrum + new geometry (near-field) + new functions (sensing) + new tools (learning)
- OFDM is not dead — it has **new roommates**; expect scenario-adaptive PHY
- Three formulas organize half the field: $P \propto N^2$ ▪ $d_R = 2D^2/\lambda$ ▪ $\text{var}(\hat{\tau}) \geq \frac{1}{8\pi^2\beta^2 \text{SNR}}$
- Signal processing remains the discipline that turns **physics into throughput**

Open problems worth a PhD

- Unified DD-domain ISAC: one waveform, provably optimal for both jobs entry: Secs. 3+6
- Near-field channel estimation & codebooks at $N > 10^3$ entry: Sec. 4
- EM-consistent RIS models (coupling, wideband) that stay tractable entry: Sec. 5
- Trustworthy neural PHY: certification, monitoring, graceful fallback entry: Sec. 7
- Energy-optimal waveform + PA co-design for sub-THz entry: Secs. 2+10
- Grant-free access at 10^8 devices/km²: CS at societal scale entry: Sec. 8

References & resources

1. ITU-R Rec. M.2160-0, "Framework and overall objectives of the future development of IMT for 2030 and beyond," Nov. 2023.
2. W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Network*, vol. 34, no. 3, 2020.
3. H. Tataria *et al.*, "6G wireless systems: Vision, requirements, challenges, insights, and opportunities," *Proc. IEEE*, vol. 109, no. 7, 2021.
4. R. Hadani *et al.*, "Orthogonal time frequency space modulation," in *Proc. IEEE WCNC*, 2017.
5. Z. Wei *et al.*, "Orthogonal time-frequency space modulation: A promising next-generation waveform," *IEEE Wireless Commun.*, vol. 28, no. 4, 2021.
6. A. Bemani, N. Ksairi, and M. Kountouris, "AFDM: Affine frequency division multiplexing for next generation wireless communications," *IEEE Trans. Wireless Commun.*, 2023.
7. O. El Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath, "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, 2014.
8. E. Björnson, J. Hoydis, and L. Sanguinetti, "Massive MIMO has unlimited capacity," *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, 2018.
9. Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, 2019.
10. M. Di Renzo *et al.*, "Smart radio environments empowered by reconfigurable intelligent surfaces," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 11, 2020.
11. F. Liu *et al.*, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, 2022.
12. C. Sturm and W. Wiesbeck, "Waveform design and signal processing aspects for fusion of wireless communications and radar sensing," *Proc. IEEE*, vol. 99, no. 7, 2011.
13. T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, 2017.
14. N. Shlezinger, J. Whang, Y. C. Eldar, and A. J. Goldsmith, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, 2023.
15. Y. Mao, O. Dizdar, B. Clerckx, R. Schober, P. Popovski, and H. V. Poor, "Rate-splitting multiple access: Fundamentals, survey, and future research trends," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 4, 2022.

Tools: MATLAB (SPT/CommT/DLT) ■ NVIDIA Sionna ■ this talk's code

Thank you!

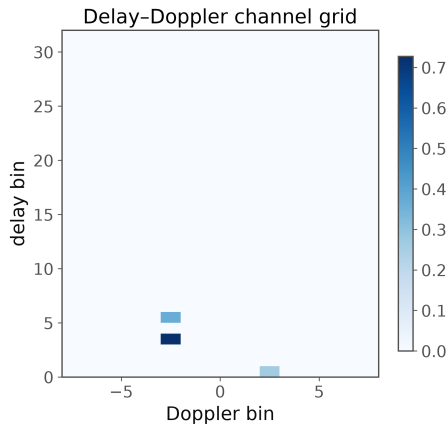
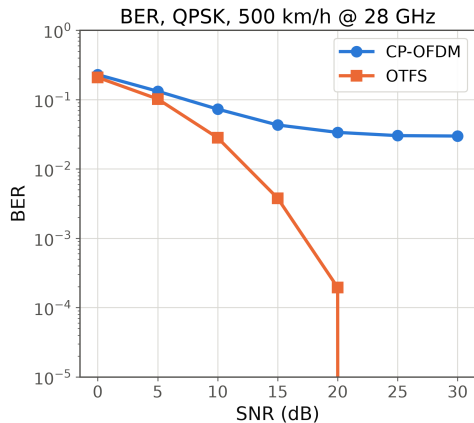
Dr. Markkandan S ■ Associate Professor
School of Electronics Engineering, VIT Chennai

Questions — now, or anytime:

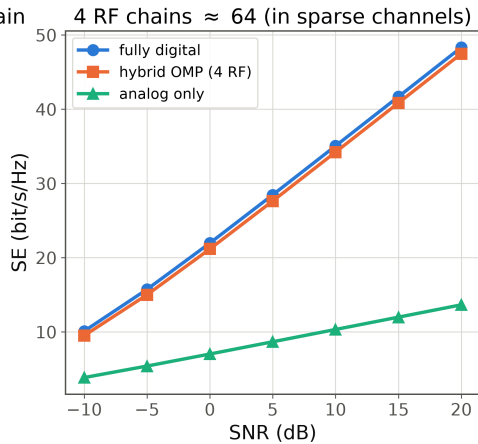
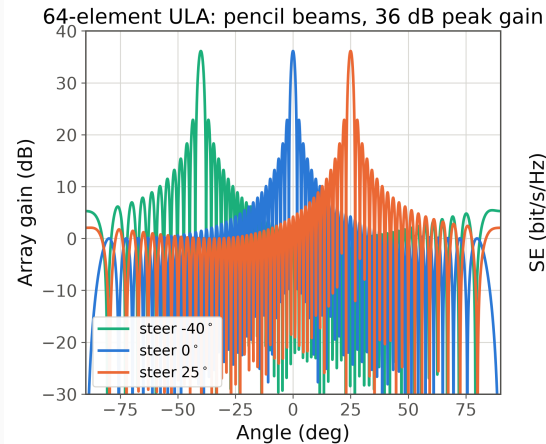
`markkandan.s@vit.ac.in`

Slides, MATLAB demos & Python verification code ship together — ask me for the archive.

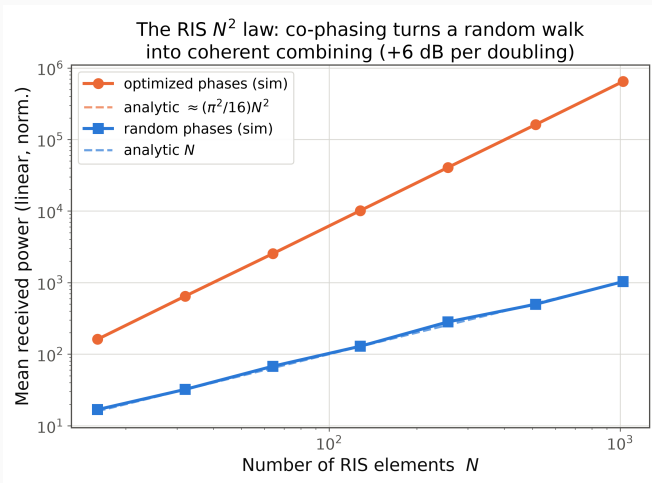
Backup A1: pre-computed result — Demo 1 (OFDM vs OTFS)



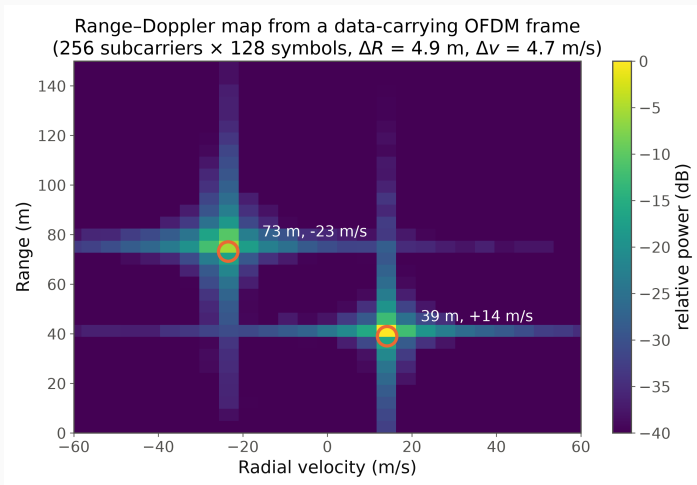
Backup A2: pre-computed result — Demo 2 (hybrid beamforming)



Backup A3: pre-computed result — Demo 3 (RIS N^2 law)



Backup A4: pre-computed result — Demo 4 (ISAC range-Doppler)



Backup A5: pre-computed result — Demo 5 (learned estimation)

